

1. Q: What is finance?

A: The art and science of managing money.

2. Q: What is financing?

A: The way a company arranges money to reach its goals.

3. Q: What is financial management?

A: Managing a company's money to make smart decisions and grow the business.

4. Q: What are the two main goals of a business?

A: Profit maximisation and wealth maximisation.

5. Q: What does profit maximisation mean?

A: Earning as much money (profit) as possible.

6. Q: What is wealth maximisation?

A: Increasing the value of the company for its owners (shareholders).

7. Q: Who is a CFO?

A: The Chief Financial Officer — the money boss of a company.

8. Q: Name two financial decisions a CFO makes.

A: Where to invest and where to get money from (like loans or shares).

9. Q: What is the financial environment?

A: The world of banks, markets, and rules that affect how companies manage money.

10. Q: What is the present value vs future value?

A: Comparing today's money to what it could be worth in the future (with interest).

11. What does a company aim to do when it talks about "wealth maximisation"?

A: Increase the value of shareholders' shares.

12. Who is in charge of managing all the money stuff in a company?

A: The CFO (Chief Financial Officer)

13. If a company takes a loan to buy machines, what is that called?

A: Financing

14. What does "Return = Risk-Free Rate + Risk Premium" mean?

A: You earn a basic amount plus extra if you take a risk.

15. Give one example of a financial institution.

A: Bank or Insurance Company

16. Explain briefly the role of the CFO.

- **Plans Finances:** The CFO forecasts how much money the company will need for daily operations and future plans. This helps avoid shortages or waste of funds.
- **Arranges Funds:** Decides how to raise money — whether by taking loans, issuing shares, or using savings. The goal is to get funds at the lowest cost with minimal risk.
- **Invests Wisely:** Chooses the best projects or assets to invest in so the company can grow. Every investment is checked for its profit potential and risks.
- **Manages Daily Money (Working Capital):** Ensures there's enough cash to pay for daily expenses like salaries, bills, and materials. Keeps the business running smoothly.

- **Handles Profits (Retained Earnings):** Decides how much profit to give to shareholders and how much to keep in the company for future growth. This reinvested profit is called retained earnings.
- **Balances Risk and Return:** Aims to get high returns but without taking dangerous financial risks. Tries to find the safest way to earn good money.
- **Maintains Liquidity:** Makes sure the company has quick access to cash or assets that can be turned into cash. This helps in paying short-term debts on time.
- **Manages Fixed Assets:** Plans for buildings, machines, and long-term tools that the company needs. Decides when and how much to invest in them.
- **Analyzes Costs and Profits:** Uses tools like break-even analysis to check if the company should make or buy something. Helps in smart decision-making to cut losses.
- **Valuates Business & Handles Restructuring:** Calculates how much the company is worth, especially during sales, mergers, or restructuring. This helps in negotiations and planning big changes.

17. What are the objectives of a business firm?

a. Profit Maximisation

- This means earning the **highest possible profit**.
- Profit is the money left after all expenses are paid.
- It is used as a measure of how efficiently the business is running.
- The idea is: the more profit a business makes, the better it is doing.

Example: If a business earns ₹1,00,000 and spends ₹60,000, the profit is ₹40,000. Maximising this profit is the goal here.

b. Wealth Maximisation

- This focuses on **increasing the value of the business** for its owners (shareholders).
- It is measured by the **market price of shares** of the company.
- Wealth maximisation is considered a **modern and better goal** than just profit.
- It considers long-term growth, risk, and sustainability.

Formula:

$$\text{Wealth (W}^0\text{)} = \text{Number of shares (N)} \times \text{Price per share (P}^0\text{)}$$

Example: If you own 100 shares and each is worth ₹50, your wealth is ₹5,000. The goal is to make the share price go up over time.

18. Why Wealth Maximisation is Better?

- It considers **risk and time value of money**.
- It focuses on **long-term benefits**, not just short-term profits.
- It ensures **shareholder satisfaction** and company reputation.

19. What is the difference between primary and secondary markets?

The primary market deals with IPOs (Initial Public Offerings) while the secondary market deals with second-hand securities.

20. Explain financial environment.

a. **Financial Institutions:** These are places that **give, manage, or move money**.

Ex. **Banks, insurance companies, housing finance companies, leasing firms, etc.**

b. **Financial Markets:** Where **buying and selling of money-related products** happens.

Two main types:

- **Money Market** – for short-term money
- **Capital Market** – for long-term money

Also includes:

- **Primary Market** – where new securities are first sold (e.g., IPO)
- **Secondary Market** – where old securities are re-sold (like a share market)

c. Financial Services: These are **services that help people and companies handle money**.

Two types:

- **Fund-based services** – involve actual money (like loans, leasing)
- **Fee-based services** – advice or help for a fee (like investment advice, insurance advisory)

d. Financial Assets: These are **money-based tools or instruments**.

Examples:

- **Direct assets** – shares, debentures, mutual fund units, etc.
- **Derivatives** – futures, options (used to manage risk)

e. Financial Regulators: These are the **rule-makers** that keep everything fair and safe.

Examples:

- **RBI** – controls money market
- **SEBI** – controls stock markets
- **IRDA** – controls insurance sector

21. How does the CFO play the role of managing cash liquidity?

- **Tracks Cash Flow Daily:** The CFO checks how much money is **coming in** and **going out**. This helps avoid any surprise cash shortages.
- **Maintains Cash Reserves:** The CFO keeps a **buffer of extra cash** ready in case of emergencies.
- **Manages Receivables and Payables:** Makes sure customers **pay on time**. Also plans **when to pay suppliers**, so the company doesn't run out of cash.
- **Avoids Unnecessary Spending:** The CFO checks all expenses to avoid overspending. Only important and planned spending is allowed.
- **Uses Short-term Loans if Needed:** If cash runs low, the CFO can arrange a **short-term loan** to keep things running smoothly without delay.
- **Keeps Good Relationships with Banks:** So that in tough times, the company can **quickly borrow cash** if needed.

22. How the CFO plays the role of managing the fixed assets of a company?

- Plans What Assets the Company Needs:** The CFO decides what fixed assets the company needs to operate or grow.
- Arranges Money to Buy Them:** Fixed assets are expensive, so the CFO plans how to pay for them — using company savings, bank loans, or issuing shares.
- Chooses Between Buy or Lease:** Sometimes it's smarter to lease (rent) machines instead of buying. The CFO compares costs and benefits to decide which option saves more money.
- Checks the Value Over Time:** As time goes on, machines and buildings lose value. This is called depreciation. The CFO keeps records of this and adjusts financial statements accordingly.
- Monitors Performance and Usage:** The CFO checks whether the fixed assets are being used efficiently. If a machine isn't working properly, it might need repair or replacement.
- Plans for Maintenance and Upgrades:** The CFO ensures money is available to maintain or upgrade old assets to avoid breakdowns and delays.

- g. **Handles Disposal or Sale of Old Assets:** When assets become outdated or are no longer useful, the CFO plans their sale, replacement, or scrapping.

23. What is Capital Budgeting?

Capital Budgeting is the process of planning where to invest money for **long-term projects** that will help a business grow.

24. Net Present Value (NPV)

NPV is a tool used in **Capital Budgeting** to decide whether a project is **worth investing in**.

NPV = PV of future money – Cost today

Example:

You invest ₹1,000 today. It gives you ₹1,200 after 1 year. The interest rate is 10%.

Answer:

First, calculate PV of ₹1,200:

$$PV = 1200 / (1+0.10)^1 = 1200 / 1.10 \approx ₹1,090.91$$

Now,

$$NPV = 1,090.91 - 1,000 = ₹90.91$$

NPV is **positive**, so this is a **profitable project**.

NPV Value	Decision
NPV > 0	Accept project
NPV = 0	Neutral
NPV < 0	Reject project

25. What is "Time Value of Money"?

Money today is worth more than money tomorrow. Because if you put ₹100 in the bank today, and it gives you 3% interest, you'll have ₹103 after one year. So, ₹100 today is **not equal** to ₹100 one year later — it's **more valuable today**.

26. You spend ₹100 on a project. In one year, it gives ₹120. If the interest rate is 10%, the present value of ₹120 is about ₹109.

$$\text{So, } NPV = ₹109 - ₹100 = ₹9 \rightarrow \text{Good Project}$$

27. You want to invest ₹1,000 in a small project. This project will give you ₹1,200 after 2 years. The bank interest rate is 10% per year. Should you invest in this project? Use NPV to decide.

$$PV = FV / (1+r)^n$$

Where:

$$FV = ₹1,200$$

$$r = 10\% = 0.10$$

$$n = 2 \text{ years}$$

$$PV = 1200 / (1+0.10)^2 = 1200 / (1.21) \approx ₹991.74$$

$$NPV = 991.74 - 1000 = -₹8.26$$

Since NPV is **negative** (–₹8.26), you **should NOT invest** in this project.

28. What does "Capital Budgeting" help you decide?

A: Whether to invest in a project or not.

29. What is the full form of NPV?

A: Net Present Value

30. If a project gives ₹1,000 after 1 year and the interest rate is 10%, what is the PV?

$$\text{A: } PV = 1000 / (1.10) = ₹909.09$$

31. What does a positive NPV mean for a project?

- A) The project will lose money
- B) The project is risky
- C) The project will make money
- D) The project should be rejected

32. You invest ₹2,000 today. The project returns ₹2,420 after 2 years. If the interest rate is 10%, what is the Present Value (PV) of ₹2,420?

- A) ₹2,000
- B) ₹2,200
- C) ₹2,000.83
- D) ₹2,420

(Hint: Use $PV = FV / (1 + r)^2$)

33. What is the Time Value of Money based on?

- A) Inflation
- B) Interest earning potential
- C) Taxes
- D) Spending habits

34. You calculate NPV and it comes out to be -₹500. What should you do?

- A) Accept the project
- B) Accept with caution
- C) Reject the project
- D) Invest more money

35. What's the future value (FV) of ₹1,000 after 3 years if interest is 5% per year?

- A) ₹1,150
- B) ₹1,157.63
- C) ₹1,050
- D) ₹1,100

(Hint: $FV = PV \times (1 + r)^n$)

36. A company wants to invest ₹5,000 in a project. The project returns ₹2,000 per year for 3 years. If the interest rate is 10%, should the company accept the project?

(Hint: You need to calculate the Present Value of each ₹2,000, then sum them up and compare with ₹5,000)

- A) Yes
- B) No
- C) Only if interest rate changes
- D) Can't decide

Answer:

→ Calculate PV for each year:

Year 1: $₹2,000 / (1.10)^1 = ₹1,818.18$

Year 2: $₹2,000 / (1.10)^2 = ₹1,652.89$

Year 3: $₹2,000 / (1.10)^3 = ₹1,502.63$

Total PV ≈ ₹4,973.70 → $NPV \approx ₹4,974 - ₹5,000 = -₹26$

37. What happens to NPV if the interest rate (discount rate) increases?

- A) NPV increases
- B) NPV decreases
- C) NPV stays the same
- D) It becomes zero

38. A project costs ₹4,000 today. It will give you ₹1,500 each year for 4 years. The interest rate is 8%. What is the total present value (rounded)?

- A) ₹5,500
- B) ₹4,989
- C) ₹4,000
- D) ₹3,812

39. Which one of the following is not a method of Capital Budgeting?

- A) Net Present Value (NPV)
- B) Internal Rate of Return (IRR)
- C) Payback Period
- D) Gross Domestic Product (GDP)

40. Why is the Present Value always less than Future Value when interest is positive?

- A) Because of inflation
- B) Because of taxes
- C) Because money today can earn interest
- D) Because people spend money early

41. A delivery startup is thinking about buying 5 electric scooters for ₹1,00,000. They expect to save ₹30,000 per year on fuel for 4 years. If the bank interest rate is 10%, should they go ahead?

- A) Yes
- B) No
- C) Only if fuel prices rise
- D) Wait for a government subsidy

42. A company can either invest in solar panels for ₹2,00,000 or continue paying ₹60,000 per year in electricity bills for 4 years. If discount rate is 10%, which option is cheaper?

- A) Invest in solar panels
- B) Keep paying electricity bills
- C) Both cost the same
- D) Depends on tax benefits

43. You run a small café. You can buy a coffee machine for ₹40,000. It will help you earn ₹15,000 extra each year for 3 years. If interest rate is 8%, what's the NPV? Should you buy it?

- A) $NPV > 0 \rightarrow$ Yes
- B) $NPV = 0 \rightarrow$ Maybe
- C) $NPV < 0 \rightarrow$ No
- D) NPV can't be calculated

44. A company considers launching a new product. It will cost ₹5,00,000 to launch. Expected return is ₹1,50,000 per year for 5 years. Interest rate is 12%. Should they launch it?

- A) Yes, profit is good
- B) No, NPV will be negative
- C) Launch if brand value increases
- D) Depends on competitors

45. Why do companies use discounting when making investment decisions?

- A) To increase profits
- B) To reduce taxes
- C) To compare future money with today's money
- D) To avoid borrowing money

NPV Formula

$$NPV = -C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_T}{(1+r)^T}$$

$-C_0 = \text{Initial Investment}$

$C = \text{Cash Flow}$

$r = \text{Discount Rate}$

$T = \text{Time}$

Interest = FV – PV

Interest Factor (Discounting Factor) = $1 / [(1 + r)^n]$

Future Value (FV)

- Compounded annually – $PV(1 + r)^n$
- Compounded semi-annually – $PV(1 + r/2)^{n \times 2}$
- Compounded quarterly – $PV(1 + r/4)^{n \times 4}$
- Compounded monthly – $PV(1 + r/12)^{n \times 12}$

Present Value (PV)

- Compounded annually – $FV / [(1 + r)^n]$
- Compounded semi-annually – $FV / [(1 + r/2)^{n \times 2}]$
- Compounded quarterly – $FV / [(1 + r/4)^{n \times 4}]$
- Compounded monthly – $FV / [(1 + r/12)^{n \times 12}]$

46. Discuss the three strategies of working capital management along with diagrammatic representation.

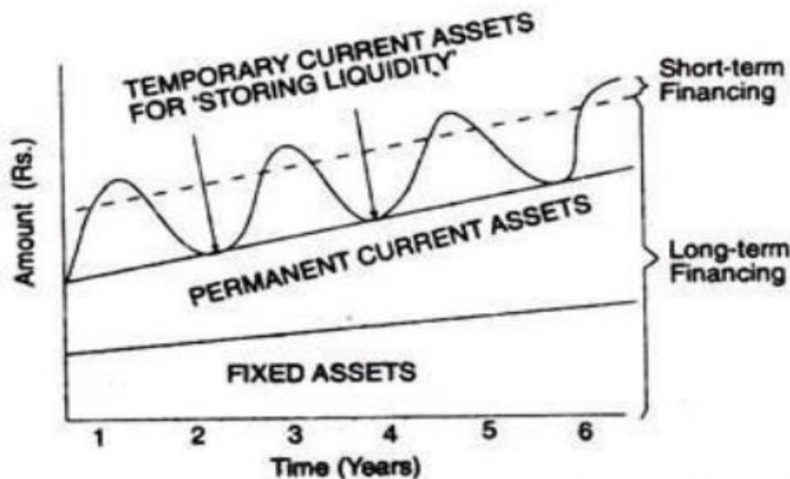
There are mainly three types of WC management strategies to choose the mix of long-term or short-term funds for financing the WC requirements of a firm. The three strategies are below:

i. Conservative strategy: In this type of strategy along with the fixed assets, permanent current assets & part of temporary current assets are financed by long-term financing. Short-term financing is used to finance the remaining part of temporary current assets.

The risk profitability trade-off in this type of strategy is low risk & low profitability.

Long-term financing >> Fixed Assets + Permanent Current Assets + Part of Temporary Current Assets

Short-term financing >> Remaining part of Temporary Current Assets



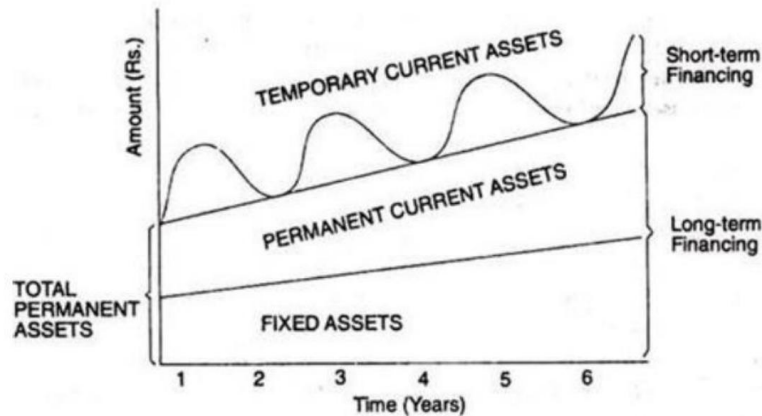
Conservative strategy

ii. Hedging Strategy: This is a balanced approach. In this type of strategy along with the fixed assets, permanent current assets are financed by long-term financing. Short-term financing is used to finance temporary current assets only.

The risk profitability trade-off in this type of strategy is moderate risk & moderate profitability.

Long-term financing >> Fixed Assets + Permanent Current Assets

Short-term financing >> Temporary Current Assets



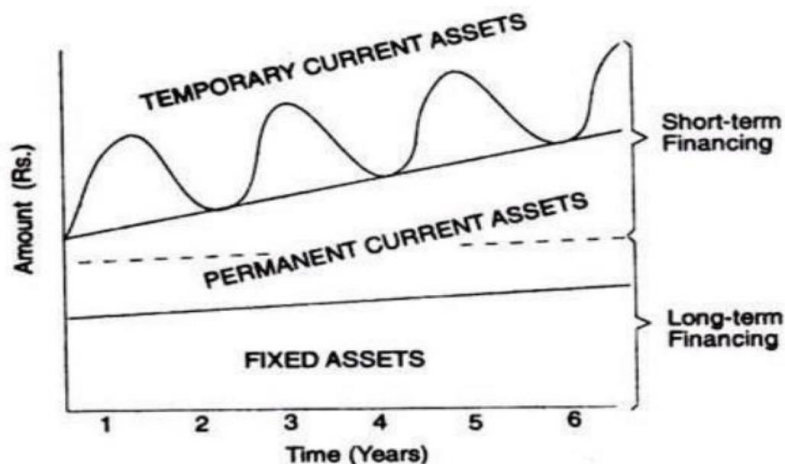
Hedging Strategy

iii. Aggressive Strategy: In this type of strategy long-term financing are used for financing fixed assets & a small portion of permanent current assets. Short-term financing is used to finance temporary current assets & major portion of permanent current assets.

The risk profitability trade-off in this type of strategy is high risk & high profitability.

Long-term financing >> Fixed Assets + Part of Permanent Current Assets

Short-term financing >> Remaining part of Permanent Current Assets + Temporary Current Assets



Aggressive Strategy

47. What is the key objective of working capital management strategies?

The objective is to decide the right mix of long-term and short-term financing to fund fixed assets, permanent current assets, and temporary current assets while balancing risk and profitability.

48. What is the main principle followed in the hedging strategy?

Assets are financed with funds of matching maturity (i.e., long-term assets with long-term funds, and short-term assets with short-term funds).

49. Why is the aggressive strategy considered risky?

Because it relies heavily on short-term financing, which is more volatile and may lead to liquidity issues.

50. What is Working Capital?

Working Capital (WC) refers to the **money a business needs to run its day-to-day operations** smoothly.

Working Capital (WC) = Current Assets – Current Liabilities

Current Assets: Cash, inventory, accounts receivable (what others owe the company)

Current Liabilities: Bills, short-term loans, accounts payable (what the company owes)

Note:-

WC Ratio = Current Assets / Current Liabilities

It gives an indication as to whether a company has enough short-term assets to meet its short-term debts.

Ratio	Meaning
2:1	Ideal: Good balance between liquidity and usage.
< 1	Bad: Company may struggle to pay debts .
> 2	Inefficient: Too much cash, not invested properly.

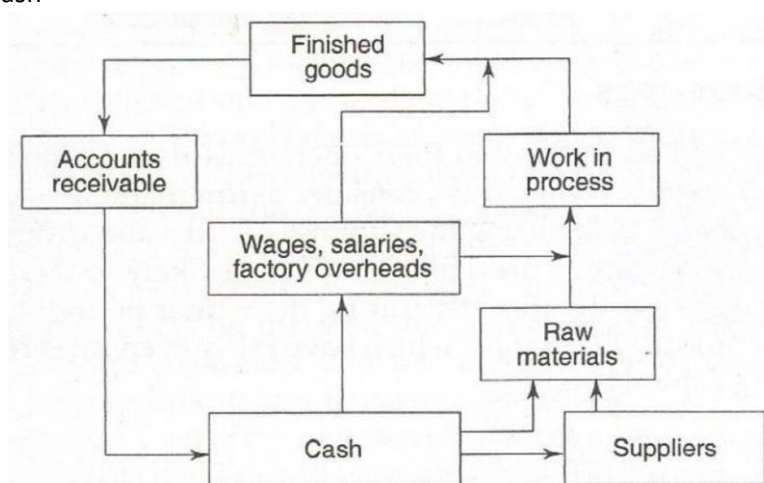
Gross WC: Total **current assets** (like cash, inventory, receivables).

Net WC: **Current Assets – Current Liabilities**

Management of Working Capital refers to the management of current assets as well as current liabilities. However, major thrust is given upon current assets. This is obvious since current liabilities are paid out of current assets.

Characteristics of Current Assets -

- **Short Life Span:** Like cash, inventory, receivables. They don't last long.
- **Quick Transformation:** These assets **change form fast**:
 - Cash → Raw Materials → Finished Goods → Credit Sales → Receivables → Back to Cash



Current Assets Cycle

From the diagram it can be inferred that cash is used for acquiring raw materials; raw materials are transformed into finished goods; finished goods are converted in to accounts receivable; & finally accounts receivable generates cash.

51. A business has: Current Assets = ₹80,000 and Current Liabilities = ₹40,000

Q: What is the Working Capital and Working Capital Ratio?

Working Capital = ₹80,000 – ₹40,000 = ₹40,000

WC Ratio = ₹80,000 / ₹40,000 = **2:1**

52. A business has: Current Assets = ₹30,000 and Current Liabilities = ₹45,000

Q: Does this business have positive or negative working capital? What is the ratio?

Working Capital = ₹30,000 – ₹45,000 = **–₹15,000**

WC Ratio = ₹30,000 / ₹45,000 = **0.67:1** (Negative working capital)

53. A business has: Current Assets = ₹1,00,000 and Current Liabilities = ₹20,000

Q: Is the company holding too much idle money? Why or why not?

WC Ratio = ₹1,00,000 / ₹20,000 = **5:1**

Too much money is idle. The company could invest more of it to grow the business instead of just keeping it.

54. Discuss the five factors influencing working capital requirements.

Here's a explanation of the five major factors influencing Working Capital (WC) requirements:

i. Nature of Business: The type of business affects how much WC is needed.

Examples:

- **Service Companies** (e.g., transport, electricity): Sell mostly for **cash**, have a **short operating cycle**, need **less WC**.
- **Manufacturing Companies** (e.g., machine tools): Sell on **credit**, have **long production cycles**, need **more WC** to buy raw materials, store goods, and wait for payments.

ii. Seasonality of Operations: Businesses with seasonal sales see fluctuating WC needs.

Example:

- A fan company:
 - **Summer:** High demand → more raw materials, production, and sales → higher WC needed
 - **Winter:** Low demand → less WC needed

iii. Production Policy: How a business plans its **production schedule** affects WC.

Two approaches:

1. **Seasonal Production:** Producing more during peak season → **higher WC needed** at certain times.
2. **Uniform Production:** Steady production throughout the year → **more stable WC needs**

Benefit:

- A steady production policy **reduces sharp spikes** in WC requirements.

iv. Market Conditions: The level of competition in the market affects WC needs.

If Competition is High:

- Must maintain large inventories to serve customers quickly.
- May need to offer credit sales.
- Results in higher WC.

If Competition is Low:

- Can keep smaller inventories.
- Can demand cash payments.
- Results in lower WC.

v. Conditions of Supply: The availability and reliability of raw material supply impacts WC.

If supply is reliable:

- Can keep small inventories → lower WC.

If supply is unpredictable or seasonal:

- Need to buy and store more materials in advance → higher WC.

55. What is meant by "swift transformation" of current assets?

Current assets quickly change form, e.g., cash → raw material → finished goods → receivables → back to cash.

56. What Is CAPM?

The CAPM is a model that describes the relationship between the expected return & systematic risk of an asset.

Note:

The formula for CAPM: $E(R) = R_f + \beta (R_m - R_f)$

where, $E(R)$ = Expected return on a security

R_f = Risk free rate

β = Beta of the security

R_m = Return of the market

$R_m - R_f$ = Risk premium

Example-(Important)

$R_f = 3\% = 0.03$

$R_m = 15\% = 0.15$

$B = 0.89$

$E(R) = ?$

Answer:

$E(R) = 0.03 + 0.89 (0.15 - 0.03) = 0.1368 = 13.68\%$

The expected return for z limited base on its level of systematic at risk is 13.68%, The company is currently priced to return 14% is higher than the would be expected return for the level of systematic risk, z limited is underpriced. Since, z limited is underpriced, one should buy the shares of z limited and earn a return higher than would be expected return for the level of systematic risk.

57. What is Risk-free Rate?

It is the return you get from **the safest possible investment**, where there's **almost no chance of losing your money**. **Examples:** Government bonds, Treasury bills etc.

58. What is Market Risk Premium?

The market-risk premium is the difference between the expected return on a market portfolio & the risk-free rate.

59. What is Beta?

Beta is a measure of a stock's volatility in relation to the overall market. By definition usually markets have a beta of 1.

Note:

A stock that swings more than the market over time has a beta of over 1 & vice versa.

High-beta stocks are supposed to be riskier but provide higher return potential & vice versa.

The beta of a stock is 0.6 which implies that a stock's movement will theoretically be about 60% of the index's movement.

By multiplying the beta value of a stock with the expected movement of an index, the expected change in the value of the stock can be determined.
Volatility is a rate at which the price of a security increases or decreases for a given set of returns.

60. Define Systematic Risk.

Systematic risk refers to the risk inherent to the entire market. Systematic risk, also known as “undiversifiable risk,” that affects the overall market, not just a particular stock or industry. This type of risk is impossible to avoid.

61. what is degree of operating leverage?

Degree of Operating Leverage (DOL) is a financial metric that measures how sensitive a company's **operating income (EBIT)** is to a change in **sales (revenue)**.

Definition:

The Degree of Operating Leverage (DOL) is the ratio of the percentage change in operating income to the percentage change in sales.

Formula:

$DOL = \% \text{Change in EBIT} / \% \text{Change in Sales}$

- **High DOL:** Indicates a high proportion of fixed costs – profits can increase rapidly with higher sales, but losses also magnify with sales decline.
- **Low DOL:** Indicates lower fixed costs – more stable earnings but less growth potential with increased sales.

Significance:

DOL helps investors and analysts understand how fixed and variable costs influence a company's profitability. Companies with high fixed costs and low variable costs tend to have higher DOLs.

Example:

If a company has:

- Sales = ₹1,00,000
- Variable Costs = ₹60,000
- Fixed Costs = ₹20,000

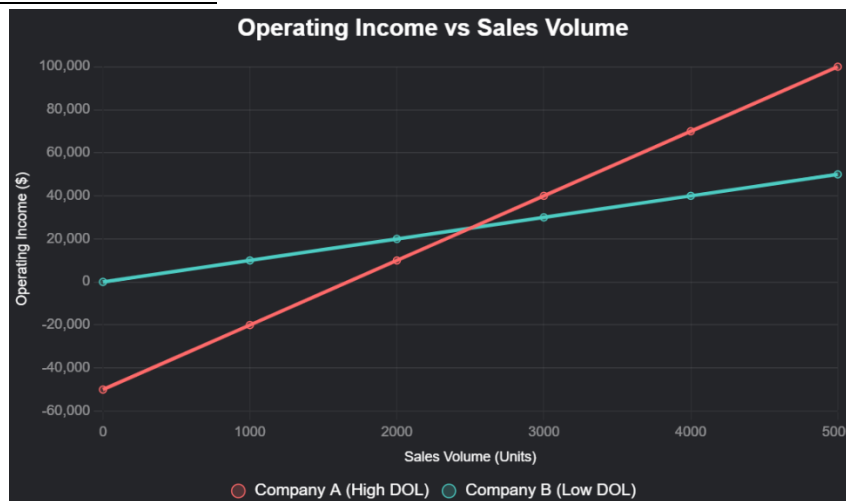
Then:

- Contribution Margin = ₹1,00,000 – ₹60,000 = ₹40,000
- EBIT = ₹40,000 – ₹20,000 = ₹20,000

$DOL = 40,000 / 20,000 = 2$

This means a **1% increase in sales** will lead to a **2% increase in EBIT**.

Pictorial representation of DOL:



62. 50000 units was purchased and sold. Selling price per unit is 25 rupees. Variable cost per unit is 15 rupees and fixed cost is 30000 rupees. Calculate degree of Leverage.

Given:

- Units sold = 50,000
- Selling price per unit = ₹25
- Variable cost per unit = ₹15
- Fixed cost = ₹30,000

Total Sales = $50,000 \times ₹25 = ₹12,50,000$

Total Variable Cost = $50,000 \times ₹15 = ₹7,50,000$

Contribution Margin = Sales – Variable Cost = $₹12,50,000 - ₹7,50,000 = ₹5,00,000$

EBIT = Contribution Margin – Fixed Cost = $₹5,00,000 - ₹30,000 = ₹4,70,000$

DOL = Contribution Margin / EBIT = $₹5,00,000 / ₹4,70,000 \approx 1.064$

This means a **1% change in sales** will result in approximately a **1.06% change in EBIT**.

63. What are the components of current assets and non-current assets.

Components of Current Assets:

- Cash and Cash Equivalents
- Accounts Receivable / Trade Receivables
- Inventory / Stock
- Short-Term Investments

Components of Non-Current Assets:

- Property, Plant, and Equipment
- Long-Term Investments
- Long-Term Loans and Advances
- Deferred Tax Assets

64. Indicate two elements having risk-free rate.

Government Treasury Bills and Government Bonds.

65. Define corporate restructuring and business evaluation.

Corporate Restructuring: Corporate restructuring is the process of reorganizing the structure, operations, or ownership of a company to improve its performance, financial health, or adapt to changing business conditions.

Business Evaluation: Business evaluation is the process of determining the economic value of a company or a business unit. It is essential for investment analysis, mergers, acquisitions, selling the business, or legal purposes.

66. What are the three types of financial decisions.

- **Investment Decision:** Deciding **where and how to allocate funds** in long-term assets or projects to generate returns over time.
- **Financing Decision:** Deciding how to raise the required funds for investments—through debt, equity, or a mix.
- **Dividend Decision:** Deciding how much profit to distribute to shareholders as dividends and how much to retain in the business for growth.

Qn:)

The details of the company's capital structure are as follows:

<u>Type of Capital</u>	<u>Book Value</u>	<u>Market Value</u>	<u>Specific Cost</u>
Equity Capital	100000	180000	15%
Preference Capital	50000	120000	12%
Debentures	60000	100000	6%
Retained Earnings	40000	—	15%

Calculate the weighted average cost of capital using —

A. Book Value as weights

B. Market Value as weights

Answer:

$$\begin{aligned}\text{Total Book Value} &= 100000 + 50000 + 60000 + 40000 \\ &= 250000\end{aligned}$$

$$\begin{aligned}\text{Total Market Value} &= 180000 + 120000 + 100000 \\ &= 400000\end{aligned}$$

A. Weighted Average Cost of Capital (WACC) using Book Value Weights —

$$\begin{aligned}\text{WACC} &= \left(\frac{100000}{250000} \times 15\% \right) + \left(\frac{50000}{250000} \times 12\% \right) + \\ &\quad \left(\frac{60000}{250000} \times 6\% \right) + \left(\frac{40000}{250000} \times 15\% \right) \\ &= (0.4 \times 15\%) + (0.2 \times 12\%) + (0.24 \times 6\%) + \\ &\quad (0.16 \times 15\%) \\ &= 6\% + 2.4\% + 1.44\% + 2.4\% \\ &= 12.24\% \quad (\text{Ans})\end{aligned}$$

(2)

B. WACC using Market Value Weights

$$WACC = \left(\frac{180000}{400000} \times 15\% \right) + \left(\frac{120000}{400000} \times 12\% \right) + \left(\frac{100000}{400000} \times 6\% \right)$$

$$= (0.45 \times 15\%) + (0.3 \times 12\%) + (0.25 \times 6\%)$$

$$= 6.75\% + 3.6\% + 1.5\%$$

$$= 11.85\% \text{ (Ans)}$$

Qn:)

ABC Ltd. is planning for the most desirable capital structure. The cost of debt (after tax) and equity capital at various levels of debt and equity mix is estimated as follows:

<u>Debt as % of total Capital employed</u>	<u>Cost of debt (%)</u>	<u>Cost of equity (%)</u>
0	10	15
20	10	15
40	12	15
50	13	18
60	14	20

Determine the composite cost of capital for each level of debt and equity and identify the optimum capital structure.

Answer:

$$WACC = \left(\frac{D}{N} \right) \times K_d + \left(\frac{E}{N} \right) \times K_e$$

where,

D = Debt proportion

E = Equity proportion (i.e., $E = 1 - D$)

N = Total capital (Debt + Equity = 100%)

K_d = Cost of debt

K_e = Cost of equity

$$\text{Debt} = 0\%$$

$$\begin{aligned}\text{WACC} &= \left(\frac{0}{100}\right) \times 10\% + \left(\frac{100}{100}\right) \times 15\% \\ &= 0 + 15\% = 15\%\end{aligned}$$

$$\text{Debt} = 20\%$$

$$\begin{aligned}\text{WACC} &= \left(\frac{20}{100}\right) \times 10\% + \left(\frac{80}{100}\right) \times 15\% \\ &= 2\% + 12\% = 14\%\end{aligned}$$

$$\text{Debt} = 40\%$$

$$\begin{aligned}\text{WACC} &= \left(\frac{40}{100}\right) \times 12\% + \left(\frac{60}{100}\right) \times 15\% \\ &= 4.8\% + 9\% = 13.8\%\end{aligned}$$

$$\text{Debt} = 50\%$$

$$\begin{aligned}\text{WACC} &= \left(\frac{50}{100}\right) \times 13\% + \left(\frac{50}{100}\right) \times 18\% \\ &= 6.5\% + 9\% = 15.5\%\end{aligned}$$

$$\text{Debt} = 60\%$$

$$\begin{aligned}\text{WACC} &= \left(\frac{60}{100}\right) \times 14\% + \left(\frac{40}{100}\right) \times 20\% \\ &= 8.4\% + 8\% = 16.4\%\end{aligned}$$

The lowest WACC is 13.8% at 40% debt. Hence, 40% debt is the optimal capital structure as it minimizes the cost of capital.

Assume that L Ltd is considering two projects namely Project X and Project Y and wants to calculate the NPV for each project. Both project X and project Y is four-year project and cash flows of both the projects for four years are given below:

Year	Project X Cash Flows	Project Y Cash Flows
1	Rs 5000	Rs 1000
2	Rs 4000	Rs 3000
3	Rs 3000	Rs 4000
4	Rs 1000	Rs 6750

The firm's cost of capital is 10% for each project and the initial investment amount is Rs 10,000. Calculate the NPV of each project and determine in which project the firm should invest.

Formula:

$$WACC = [E/V * R_e] + D/V * [R_d (1 - T)]$$

Where:

E = Market value of the firm's equity (market cap)

D = Market value of the firm's debt

V = Total value of capital (equity plus debt)

E/V = Percentage of capital that is equity

D/V = Percentage of capital that is debt

R_e = Cost of equity

R_d = Cost of debt

T = Tax rate

1) The following is the capital structure of a company.

Source	Amount (Rs.)	Before tax cost (%)
Equity capital	3,00,000	15
Retained earnings	2,00,000	13
Preference capital	1,50,000	16
Debentures	3,50,000	12

Assume tax rate at 30%. Compute weighted average cost of capital.

Answer:

tenure = 4 years
 cost of capital = 10%
 initial investment = ₹ 10,000

for project X,

Formula, $NPV = -C_0 + \frac{C_1}{1+r} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_n}{(1+r)^n}$

for X

$$NPV = -10000 + \frac{5000}{1+0.1} + \frac{4000}{(1+0.1)^2} + \frac{3000}{(1+0.1)^3} + \frac{1000}{(1+0.1)^4}$$

$$= -10000 + 4545.45 + 3305.78 + 2253.94 + 683.01$$

$$= 788.18$$

for Y

$$NPV = -10000 + \frac{1000}{1+0.1} + \frac{3000}{(1+0.1)^2} + \frac{4000}{(1+0.1)^3} + \frac{6750}{(1+0.1)^4}$$

$$= -10000 + 909.09 + 2479.33 + 3005.25 + 4610.34$$

$$= 1004.01$$

∴ Since NPV of project Y is higher than project X hence project Y is accepted and X is rejected.

Answer:

Source	Amount (Rs.)	Cost (Before Tax %)
Equity Capital	3,00,000	15%
Retained Earnings	2,00,000	13%
Preference Capital	1,50,000	16%
Debentures	3,50,000	12%

Tax rate = 30%
 Total Capital = 3,00,000 + 2,00,000 + 1,50,000 + 3,50,000 = 10,00,000
 After-tax cost of debt = 12% × (1 - 0.30) = 8.4%
 Equity Capital:
 (3,00,000 / 10,00,000) × 15 = 0.3 × 15 = 4.5%
 Retained Earnings:
 (2,00,000 / 10,00,000) × 13 = 0.2 × 13 = 2.6%
 Preference Capital:
 (1,50,000 / 10,00,000) × 16 = 0.15 × 16 = 2.4%
 Debentures:
 (3,50,000 / 10,00,000) × 8.4 = 0.35 × 8.4 = 2.94%
 WACC = 4.5 + 2.6 + 2.4 + 2.94 = 12.44%